

DNS: IF (IP/hostname resolved): Just IP Else: <nameserver, IP>	NAT IF same IP:port send with established outfacing IP:Port. WAN side address: Router Interface IP, Port++. LAN: IP and port.	SPLIT HORIZON/POISON REVERSE: Does neighbor use focus to get to node on first hop. Yes, infinity. No, count.
Distance Vector: when computing use full weights Create Node x Node table Initially fill with immediate nodes (1 edge) Fill with distance to node starting at 2 edges now	LINK STATE ROUTING: Run Dijkstra's on a node (To, Distance, Using) Remember it's weight from starting so make sure to count up (B,0,-) (E,1,E) (A,2,A) Added E (B,0,-) (E,1,E) (A,2,A) (C,7,E) (F,4,E) E explore (B,0,-) (E,1,E) (A,2,A) (C,7,E) (F,4,E) Added A (B,0,-) (E,1,E) (A,2,A) (C,7,E) (F,4,E) (D,5,A) Aexp	Vectors A would receive from neighbors D: A B C D E F Inf Inf Inf 0 Inf Inf B: A B C D E F Inf 0 5 Inf 1 4 C: A B C D E F 7 5 0 10 4 1 ARP Format: :SRC Type Port DST Port

BGP Routing: Customer-Provider Customer needs to be reachable from everyone Provider tells all neighbors how to reach the customer Customer does not want to provide transit service Customer does not let its providers route through it Peer-Peer Peers exchange traffic between customers AS exports only customer routes to a peer AS exports a peer's routes only to its customers Traffic shaping controls the rate of sending by buffering excess incoming traffic, while traffic policing drops any excess.	Signaling: NRZ(Non return to zero) - High 1, Low 0 NRZI(NRZ Inverted) - Change 1, Maintain 0 Manchester - NRZ (xor) clock - 50% efficiency Asynchronous: send these training bits every time Synchronous: big up front(preamble) send for long period (trailer) - (i.e. NRZ NRZI Manchester) Formulas: $RTT = 2 * (distance) / (speed\ of\ light)$ $Bandwidth-delay = (bandwidth) * (RTT/2)$ $Time\ to\ transmit\ all\ data = (data\ to\ transfer) / (bandwidth)$ $Total\ Time\ to\ get\ all\ data = (time\ to\ transmit\ all\ data) + (RTT/2)$ Bandwidth: Power of 10 (1 Mbps: $1 * 10^6$) File Size: Power of 2 (2^{20} bytes is 1 MB)
Scheduling & QOS - Treat flows individually, treat size as time - If weighted, divide size by weight	Speed of Light: ($3 * 10^8$) Distance: meters Bandwidth: bits/second

HW 1: (ASEs do not wish to expose their internal topology to other ASEs)

- Sentinel: 101 1111 10
- $2d + 1 = \text{hamming distance}$, $2d = \text{detection}$, $d = \text{correction}$, one bit parity has hamming distance of 2
- CRC: Pad data with $\text{len}(\text{CRC}) - 1$ 0s, divide by CRC replace 0s with remainder
- Learning Bridges: If a packet is sent by a host, bridges will know how to get to that host, else transmit to everyone
- Spanning Tree: take shortest path to B1, if tie pick smaller router number

Naming

- DHCP** - Automates host boot up, given a MAC assigns IP
 - On Boot up : Host: "DHCP discover" -> DHCP server: "DHCP offer" -> Host: request -> DHCP server: IP, "DHCP Ack"

Link State Routing AND Distance Vector Routing

- Forwarding Options: Source Routing vs Virtual Circuits vs Destination Based Forwarding (Forwarding Tables)
- Routing Options: Link State Routing vs Distance Vector vs Path Vector
- Link State Routing - Reliable Flooding (tell all routers what you know) then Path Calculation (Dijkstra's)
 - Link State Converges quickly on topology failure, but causes flood of updates(**Transient Disruption**)
 - Examples: IS-IS, OSPF(Open Shortest Path First)
 - Strengths: Loop Free, Messages are small, converges quickly, guaranteed convergence
 - Weakness: Must flood data across entire network & Must maintain entire topology
- Distance Vector: Each Router knows own address and cost to reach direct neighbors
 - $dx(y) = \min \{c(x,v) + dv(y)\}$ - Iterative, asynchronous, distributed
 - Problem: Counting to infinity if one edge goes down
 - Solutions: Hold Downs, Loop Avoidance, Split Horizon w/ Poison Reverse (only on loops size 2)
 - Examples: RIP(Routing Information Protocol) & EIGRP: (ensures loop freedom)
 - Weaknesses: Weak at adapting to changes out of the box

Interdomain Routing

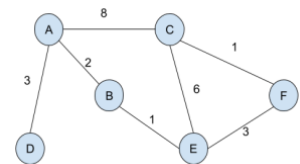
- Exterior Gateway Protocols (EGPs): Only exchange reachability information (not specific metrics)
- Path vector: send the entire path for each destination - Node can easily detect/discard a loop/keep local policies
- Border Gateway Protocol (BGP): Establish Session -> Exchange all active routes ->Exchange incremental updates
 - Attributes: AS path, next hop, ORigin, Local pref(rankings), multi exit discriminator, community
 - Problems: Instability: not guaranteed to converge, Performance, Scalability

Router Design

- Control pane(slow, handles firewall, routing table, admission control) vs Data Pane (fast, per packet, handles forwarding, switching, output scheduling)
- Buffer management:** Which packets to drop? Vs **Scheduling:** which packets to transmit next
- AQM(Active Queue Management):** Engage the router in the congestion control by detecting overloading b/c router can see actual queue occupancy

RED - If avg < min-th do nothing - If avg > max-th, drop packet - Else drop/mark packet proportional to queue length

Token Bucket - r (average rate token gets dropped), b(bucket depth), R(maximum rate tokens leave - drain rate)



NASA launched a mars rover, called Curiosity, designed to study the planet's climate and geology. Curiosity has a few communications system. It can communicate with the Mars Reconnaissance Orbiter (MRO) at a speed of 2 Mbps with its UHF band radio. Assume that any communication done between Curiosity and the MRO is done when they are at a constant distance of 500 km (unrealistic, but just use it).

a. Calculate the minimum RTT and the bandwidth-delay product for the link between Curiosity and the MRO. Assume this is a one-way link.
 $RTT = 2 * (distance) / (speed\ of\ light) = 2 * (500 * 10^3) / (3 * 10^8) = 3.33\ ms$
 $Bandwidth-delay = (bandwidth) * (RTT/2) = (2 * 10^6) * (3.33 * 10^{-3}) / 2 = \text{about } 3333\ bits$

b. Let us assume that the space probe sends 10MB of data to the MRO when it can. What would be the total time for the MRO to get the entire 10MB of data? (ie. the time taken for a single 10MB data transfer)
 $Data\ to\ transfer = 10MB = 80 * 2^{20}\ bits$
 $Time\ to\ transmit\ all\ data = (data\ to\ transfer) / (bandwidth) = (80 * 2^{20}) / (2 * 10^6) = \text{about } 41.943\ s$
 $Total\ Time\ to\ get\ all\ data = (time\ to\ transmit\ all\ data) + (RTT/2) = (41.943) + (.00166) = 41.94466\ s$
 I will grade the total time up to the precision of whatever decimal place is given.

c. If the MRO is only in range of Curiosity for 8 minute intervals, how much data can be transferred from Curiosity to the MRO in one interval?
 $Max\ data\ transfer = (bandwidth) * (time\ of\ interval) = (2 * 10^6) * (60 * 8) = 960 * 10^6\ bits = (960 * 10^6) / (8 * 2^{20})\ MB = \text{about } 114.44MB$
 Assume the speed of light is $3 * 10^8\ m/s$

"DHCP

state for

works

Links And Signaling

- **Integrated Services** (guarantees end-to-end delivery) vs **Differentiated Services** (motivated by economics)
- **Modem**: takes input stream of bits, modulates some physical media to send data, demodulates the signal to get bits.
- **Signal**: is some form of energy (light, voltage, etc) travels on **Channel** (a physical medium that conveys energy)
- **Bandwidth-Limited**: refers to the the range of frequencies the channel will transmit
- Power **attenuates** over distance
- Different frequencies have different response (**distortion**) AKA attenuate differently.
- **Coaxial Cable**: copper core, insulation, braided outer conductor, outer insulation (**twisted pair**)
- **Fiber Optics**: comes in two modes (multimode and single mode - longer but stricter)
- **Spectrum Allocation**: a policy that forces the radio spectrum to be allocated like a fixed spatial resource
- **Baseband modulation**: means to send the "bare" signal so all signals fall in the same frequency range
- **Broadband modulation**: shifts signal to carrier frequency and sends it out

Weighted Fair Queuing implements max-min fairness

Modulation/Clock Recovery

- **Delay(seconds), Overhead(seconds/cycles), Error rate(probability)**
- **Amplitude Shift Keying** (louder: 1, quieter: 0) vs **Frequency Shift Keying** (high: 1, low: 0) vs **Phase SK(Shift phase)**
- Bandlimited channels cannot respond faster than some maximum frequency f , if attempted (**intersymbol interference**)
- **Nyquist Limit** tells us In a channel bandlimited to f , we can send at maximum $2f$ symbol (**baud rate**).
- Baud rate is the speed at which symbols can be communicated across a channel; bit rate depends on the information rate of each symbol. I.e., if each symbol corresponds to more than one bit, bit rate will exceed baud rate.
- Channel Capacity: $C < 2 (\text{bandwidth}) \log (\text{base } 2) (M)$ Where M is levels in voltage
- Real channels are **noisy** and thus limits how many levels we can send
 - Number of bits per symbol: $\log (\text{base } 2) (S/2N)$
 - S = Signal Power, $0V-3B$, N = Noise, $0.5V$, $\log(\text{base } 2) (3/1) = 1.58$ bits per symbol
- Shannon's Law - upper bound on any channel's performance: **Transmit Time: $T = M/R + D$**
- **Bandwidth-Delay Product: $R * D$** - Refers to how many bits can be "stored" in transit, fill the channel.
- So long as the receiver samples at $2f$ All is good, based off of the **Nyquist formula**
- determine when to start sampling by sending a few training bits (**initial training bits**)
- We also need to combat clock drift as signal proceeds, so we use **transitions** to keep clocks synced up
- Congestion control prevents overrunning buffers in the network, while flow control prevents overflowing the receiver's

Collision: Window-based flow control is simple to implement than rate based and does not require fine-grained clocks, but leads to more bursty behavior than rate-based flow control.

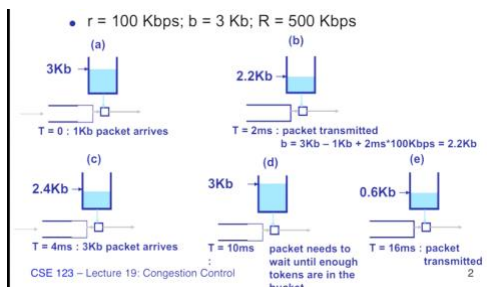
Media Access Control

- **4B/5B** - is a byproduct of manchester trying to prevent baseline drift by making sure 1 and 0s occur frequently(80%)
- How to share Medium with multiple nodes **Frequency Division Multiple Access (FDMA)** vs **Time Division Multiple Access (TDMA)** vs **Code Division Multiple Access (CDMA)**
- Guaranteed bandwidth but aren't well suited for random access usage bc wasteful when not used
- **Aloha** - retransmit after a random delay vs **Slotted Aloha** backoff in slots instead of seconds (37%)
 - Requires **time synchronization** between hosts
- **Carrier Sense Multiple Access (CSMA)** - sending and listening at the same time
 - **Non-persistent CSMA**: give up, or send after some random delay vs **1-persistent CSMA**: send as soon as channel is idle. Vs **P-persistent CSMA**: send the packet with probability p where $(p * \text{number of packets}) < 1$
- **CSMA/CD - collision detection** - Requires a minimum frame size
 - In order for collision detection to work, one of the hosts must be sending for $2 * (\text{the propagation delay})$ amount
 - works well at load but collisions are a problem when load is high
- **Contention Free Protocols** - (Polling vs Token Passing - **Fiber Distributed Data Interface (FDDI)**)
 - can make bandwidth guarantees, but complex and fragile

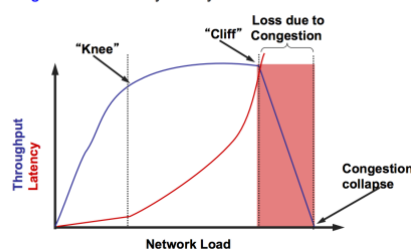
Ethernet: Ethernet won because it was cheap and easy to scale

802.11 (Frame Format) - Start Frame Delimiter + Signal(describes data rate of the payload) + Data) - uses CSMA

- **Hidden Terminal Problem [A-----B-----C]**
 - B can communicate with both A and C. A and C cannot hear each other. A transmits to B, C cannot detect the transmission using the **carrier sense** mechanism. Collision will occur at B.
- Wifi uses **half duplex** so it can't detect collision, uses **collision avoidance** with **carrier sense**
- **Collision Avoidance - RTS/CTS (MACA)** - Ensures C hears A when B rebroadcasts - If RTS packet collide, backoff in proportion to contention window
- **Distributed Coordination Function (DCF) - CSMA/CA**
 - Uses a **Network Allocation Vector (NAV)** to implement collision avoidance
 - "The stations listening on the wireless medium read the Duration field and set their NAV, which is an indicator for a station on how long it must defer from accessing the medium."
 - Combined with RTS/CTS Avoids hidden terminal problem



- **Congestion avoidance**: try to stay to the left of the knee
- **Congestion control**: try to stay to the left of the cliff



a. Host A sends a packet to B
Step 1- A sends an ARP request on port 1 for 78.64.1.1
Step 2- R1 responds on port 1 with E6-E9-00-17-BB-4B
Step 3- A sends the packet on port 1 with source IP 78.64.1.5, destination IP 78.64.3.5, source MAC address 74-29-9C-E8-FF-55, and destination MAC address E6-E9-00-17-BB-4B
Step 4- R1 sends an ARP request on port 2 for 78.64.2.2
Step 5- R2 responds on port 1 with 1A-23-F9-CD-06-9B
Step 6- R1 sends the packet on port 2 with source IP 78.64.1.5, destination IP 78.64.3.5, source MAC address CC-49-DE-D0-AB-7D, and destination MAC address 1A-23-F9-CD-06-9B
Step 7- R2 sends an ARP request on port 2 for 78.64.3.5
Step 8- B responds on port 1 with 2A-1A-CD-DE-E9-74
Step 9- R2 sends the packet on port 2 with source IP 78.64.1.5, destination IP 78.64.3.5, source MAC address 88-B2-2F-54-1A-0F, and destination MAC address 2A-1A-CD-DE-E9-74